

FoodSaver

Project Overview, CIS422 Human-Computer Interaction

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Introduction

Food waste is a persistent sustainability challenge, and the issue is especially pronounced at the household level in Saudi Arabia. Common causes include weak inventory tracking, forgotten expiration dates, and purchasing decisions that are not consistently aligned with what is already available at home. FoodSaver is a mobile application prototype designed to address this gap by giving users a simple, visual way to track food, receive timely reminders, and turn what they already have into meals.

Project Description

FoodSaver is a smart home-based mobile application for reducing food waste and managing household food stock. The application uses a calendar-style view for tracking expiration dates, checkbox-based ingredient selection instead of manual typing, and a recommendation engine that suggests recipes based on ingredients the user already owns. A community section allows users to share and rate recipes, and a donation feature connects surplus food with local charities.

The application was designed around four groups of food decision makers: parents, homemakers, university students living independently, and other family members who regularly handle groceries and cooking at home.

Core Features

Feature	Description
Inventory tracking	Food items are logged with expiration dates and displayed in a calendar-style view.
Expiration reminders	Automatic notifications are sent before items expire, reducing reliance on memory.
Recipe suggestions	Recipes are recommended from checkbox-selected ingredients rather than free text entry.
Community and donation	Users can share and rate recipes, and donate surplus food to nearby charities.

Research Methodology

The project began with a competitive analysis of four existing systems: NoWaste and FridgePal for inventory management, Yummly for recipe recommendation, and Too Good To Go for surplus food redistribution. No single system combined all of these capabilities, which shaped the scope of FoodSaver.

Primary research combined two methods. Ten semi-structured, face-to-face interviews were conducted with adults actively involved in household food management, each lasting between ten and twenty minutes, with written informed consent obtained in advance. Findings from the interviews were then used to design a structured online survey, completed by thirty-five participants, to quantify the patterns identified during the interviews. A task analysis identified six key user tasks: logging inventory, scheduling expiration, receiving smart notifications, discovering recipes, sharing within the community, and expert recipe validation.

Key Research Findings

Finding	Result
Responsible for grocery shopping and cooking	82.8% of survey respondents
Do not consistently check expiration dates	57.1% of survey respondents
Consider expiration reminders the most useful feature	54.3% of survey respondents
Discard food without noticing it had expired	48.6% of survey respondents
Want recipe suggestions based on available ingredients	51.4% of survey respondents
Prefer checkbox selection over typing ingredients	34.3% of survey respondents

These findings directly shaped the design of FoodSaver, in particular the priority given to expiration reminders and to checkbox-based ingredient selection over manual text entry.

Design Process

The design followed a User-Centered Design (UCD) framework, chosen because food waste is closely tied to everyday user behavior rather than fixed assumptions. Fogg's Behavior Model (FBM) was layered on top of UCD as a persuasive design tool, ensuring that each core feature addressed motivation, ability, and a timely trigger together. For example, reward badges support motivation after a donation, checkbox-based entry keeps the ability barrier low, and push notifications act as the trigger before food expires.

Phase	Focus	Outcome
Phase 1	Pre-design analysis	Competitive analysis of existing systems and identification of key user tasks.
Phase 2	User research	Interviews, survey, and task analysis that informed the initial paper prototype.

Phase	Focus	Outcome
Phase 3	Design refinement	Revised interfaces addressing usability issues, leading to a high-fidelity Figma prototype.
Phase 4	Usability testing	Think-Aloud sessions evaluated against Nielsen's usability heuristics.

Usability Testing

Three participants representing key target segments, two university students and one housewife, completed Think-Aloud usability sessions covering the donation flow, the inventory and calendar screens, and the recipe suggestion flow. All three sessions were completed successfully, with task times ranging from about twenty to forty-five seconds depending on the flow being tested.

Area Tested	Strength Identified	Issue Identified
Donation flow	Reward badges made the donation process feel rewarding and motivated repeat use.	No review step before the donation is submitted, so items are not double-checked beforehand.
Inventory and calendar	Navigation between the Home and Calendar screens was intuitive overall.	The grey expiration indicator was mistaken for inactive information, and past dates on the calendar caused confusion about whether they were selectable.
Recipe suggestions	Ingredient tags on the recipe screen reduced the need to remember earlier selections.	The app does not show which ingredients are missing for a recipe, limiting meal planning.

These findings were carried forward into a set of heuristic-based improvements, including a standardized onboarding flow, a faded style for past calendar dates, clearer error recovery messages, color-coded indicators for missing ingredients, and a bulk mark-as-read option for notifications.

Tools and Technology

The prototype was built in Figma, progressing from an initial paper prototype through design refinement to a high-fidelity interactive prototype of more than fifteen screens. Interview data was collected through notes and, with participant permission, audio recording, while the survey was distributed and collected through Google Forms.

Conclusion

The FoodSaver prototype demonstrates how a combination of user-centered design and persuasive design theory can make household food management easier and more immediate. The evaluation results show that the core features, an expiration calendar, a checkbox-based inventory system, ingredient-driven recipe recommendations, and a donation module, address real gaps identified through

the project's research. Suggested directions for future work include AI-assisted consumption analysis, barcode scanning for faster item entry, and deeper integration with grocery stores and charities.